SE3002 - Software Quality Engineering

Assignment 2: Final Testing Report - Student Course Registration System

GitHub repository: https://github.com/Muhammad-SE-Eng/SQE-Assignment-2
Contributors and reviews: See merged pull requests.

1. Black-box design: ECP, BVA, decision table, optimization

We wrote the expected results from REQUIREMENTS.md (FR-01 to FR-15) before looking at registration.py. The black-box file tests/test_blackbox.py has 18 ECP tests, 20 BVA tests and 9 decision table tests, plus tests for registration and withdrawal state changes (59 tests in total). The design sheets are in docs/blackbox_test_design.xlsx and docs/decision_table.xlsx.

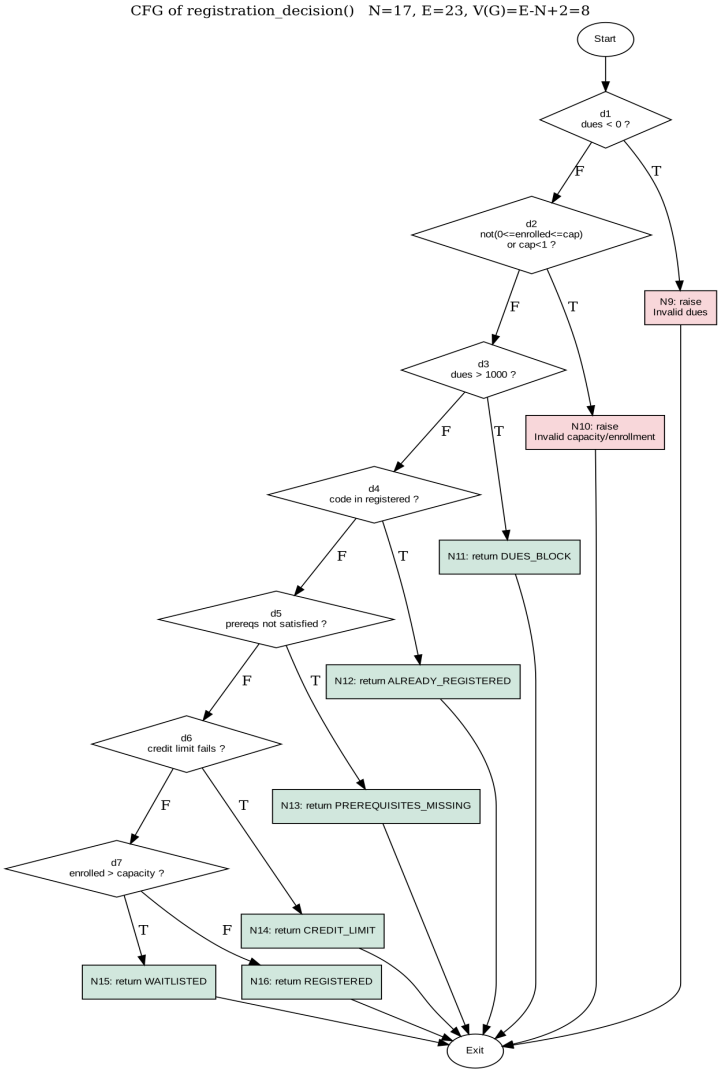
ECP: dues (zero, positive, negative), prerequisites (none, all, some, none completed), already registered, seats free or full, course credits valid or invalid, current credits invalid, capacity/enrolled invalid. **BVA:** current credits 0, 18 (and -1, 19); course credits 1, 4 (and 0, 5); total credits 18 vs 19 (14+4, 15+3, 17+1 vs 15+4, 16+3, 18+1); enrolled = capacity-1, capacity, capacity+1, -1; capacity 0 and 1; dues -1, 0, 1.

Rule	C1 dues>0	C2 already reg.	C3 prereq missing	C4 credit >18	C5 full	Outcome
DT-01	T	-	-	-	-	DUES_BLOCK
DT-02	F	T	-	-	-	ALREADY_REGISTERED
DT-03	F	F	T	-	-	PREREQUISITES_MISSING
DT-04	F	F	F	T	-	CREDIT_LIMIT
DT-05	F	F	F	F	T	WAITLISTED
DT-06	F	F	F	F	F	REGISTERED

Priority combinations (all conditions below the first applicable one set to True) are tested by DT-01 to DT-04 and by DT-07 (dues+registered), DT-08 (registered+prereq), DT-09 (prereq+credit), DT-10 (credit+full) and DT-11 (invalid dues checked before invalid capacity).

Optimization (A5). A case was written only once when two techniques used exactly the same input and expected result. For example the normal eligible student is ECP-01, ECP-04, ECP-09, BVA-15 and DT-06 together, and the full course is ECP-10, BVA-12, DT-05 and AC-05. Cases with the same output but different input (dues 500 and dues 1) were kept separate. Each technique id is still written in the test comment so it can be traced.

2. White-box analysis of registration_decision()



Cyclomatic complexity. N = 17 (Start, 7 decision nodes, 8 return/raise nodes, Exit), E = 23, so V(G) = E - N + 2 = **8**. Method A (each compound predicate = one decision node): 7 decisions + 1 = 8, matching V(G). Method B (each atomic condition of the compound predicate d2 counted separately, chained comparison = 2 atoms): 9 decisions + 1 = 10; V(G) of the condition-level graph is also 10.

Independent basis paths (8). P1 d1=T; P2 d1=F,d2=T; P3 d3=T; P4 d4=T; P5 d5=T; P6 d6=T; P7 d7=T (WAITLISTED); P8 all False (REGISTERED). Tests: test_path1 to test_path8. P7 is **infeasible in the supplied code** because d2 guarantees enrolled <= capacity; this is defect D4. After the fix (>=) P7 is feasible and tested by test_ecp10.

MC/DC for d2 = not(a and b) or c, with a: 0 <= enrolled, b: enrolled <= capacity, c: capacity < 1 (N = not evaluated, due to Python short-circuiting):

Test	a	b	c	D	(enrolled, capacity)
MCDC-1	F	N	N	T	(-1, 30)
MCDC-2	T	F	N	T	(31, 30)
MCDC-3	T	T	F	F	(5, 30)
MCDC-4	T	T	T	T	(0, 0)

Independent-effect pairs: a - (MCDC-1, MCDC-3); b - (MCDC-2, MCDC-3); c - (MCDC-3, MCDC-4). Four tests (n+1) achieve MC/DC. Condition coverage on the other predicates is trivial (single atomic conditions). pytest-cov does not measure condition coverage or MC/DC, so these were calculated manually.

3. Coverage before and after white-box additions

Stage	Tests	Statements	Branch	Total cover	Uncovered
Initial (black-box only, original code)	59 (42 pass, 17 fail)	57, miss 3	26, partial 3	93%	line 51 (DUES_BLOCK return), 59 (WAITLISTED return), 71 (waitlist append)
Final (+ white-box, original code)	76 (58 pass, 18 fail)	57, miss 2	26, partial 2	95%	lines 59 and 71 - cannot be reached because of D4
After defect fixes (regression)	76 (76 pass)	54, miss 0	26, partial 0	100%	none

Tests added in test_whitebox.py: test_wb01 (dues over 1000, covers line 51), test_wb02 (dues = 1000), test_wb03 to test_wb06 (other branches of register_course, check_credit_limit, prerequisites_satisfied and withdraw_course), test_path1 to test_path8 and test_mcdc1 to test_mcdc4. Lines 59 and 71 (WAITLISTED return and adding to the waitlist) cannot be covered with the given code because of defect D4, so we did not force them; after the fix they are covered by the black-box tests (100%). Line 51 was missed at first only because no black-box test used dues above 1000. Even the first 93% found all 5 defects, so a high coverage number does not mean the code is correct.

4. Defects and fix status

ID	Requirement	Defect	Failing tests	Issue / status
D1	FR-01, FR-07	check_credit_limit uses < 18 (total of 18 rejected)	bva23, bva24, bva25, bva34	Fixed (PR #8)
D2	FR-03, FR-06	prerequisites checked with intersection (any) not all	ecp21, ecp06	Fixed (PR #8)
D3	FR-04	dues block only when > 1000; must be > 0	ecp02, bva16, dt01, dt07, failed_dues_no_change, wb02	Fixed (PR #8)
D4	FR-08, FR-12	waitlist test enrolled > capacity unreachable; full course returns REGISTERED	ecp10, bva21, waitlist_adds_student, waitlist_no_duplicate	Fixed (PR #8)
D5	FR-13	withdraw_course does not decrement course.enrolled	withdraw_success, register_then_withdraw	Fixed (PR #8)

Initial failing runs were saved (evidence/blackbox_initial_run.txt, evidence/coverage_initial.txt, and the first GitHub Actions run) before any fix. Fixes are on the defect-fixes branch; after the fixes all 76 tests pass. Expected values were never edited to match the code.

5. Repository, pull requests and contributions

Item	Link / detail
Repository	https://github.com/Muhammad-SE-Eng/SQE-Assignment-2
PR #1 blackbox-testing	https://github.com/Muhammad-SE-Eng/SQE-Assignment-2/pull/1
PR #7 whitebox-testing	https://github.com/Muhammad-SE-Eng/SQE-Assignment-2/pull/7
PR #8 defect-fixes	https://github.com/Muhammad-SE-Eng/SQE-Assignment-2/pull/8
GitHub Actions: failing run before fixes	See repository Actions tab
GitHub Actions: passing run after fixes	See repository Actions tab

Member	Contribution
Ghulam Muhammad (24P-3116)	Black-box and white-box testing, defect fixes, regression testing and final report.
Vinod Kumar (24P-3012)	Group member; collaborative work on the Student Course Registration System.